MODULE 6: SEMICONDUCTOR DEVICES AND APPLICATIONS

(CO: 6)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with VIT BEEE102L Official Syllabus, VIT Department Question Bank, and VIT FAT Past Papers

Scope of This Textbook (All Syllabus Topics): (1) Intrinsic and extrinsic semiconductor physics, drift and diffusion current, (2) P-N Junction Diode: Forward and reverse bias V-I characteristics, dynamic resistance, (3) Zener diode: breakdown mechanisms (Zener vs Avalanche) and shunt voltage regulator design, (4) Power rectifiers: Half-wave, Center-tapped, and Full-wave bridge rectifiers (efficiency, ripple factor, PIV), (5) Bipolar Junction Transistor (BJT): CE input/output characteristics, operating regions, and Transistor switch, (6) Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET): Enhancement vs Depletion mode, and (7) Operational Amplifiers (Op-Amps): Inverting, non-inverting, summing, and differential configurations.

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1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 6

In VIT Final Assessment Tests, Module 6 carries 20 to 25 marks:

1. **Question Type A (10 Marks):** Full-Wave Bridge Rectifier: Derivation of DC load current $I_{\text{dc}} = \frac{2I_m}{\pi}$, RMS current $I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$, rectification efficiency $\eta = 81.2\%$, and ripple factor $\gamma = 0.482$.
2. **Question Type B (10 Marks):** Zener Diode Voltage Regulator: Calculation of series limiting resistor R_s, maximum and minimum Zener current ($I_{\text{Z,min}}, I_{\text{Z,max}}$), or Op-Amp closed-loop gain derivations.

Device / Parameter	Half-Wave Rectifier	Full-Wave Bridge Rectifier
DC Load Voltage (V_{dc})	$\frac{V_m}{\pi} \approx 0.318V_m$	$\frac{2V_m}{\pi} \approx 0.636V_m$
RMS Load Voltage (V_{rms})	$\frac{V_m}{2} = 0.5V_m$	$\frac{V_m}{\sqrt{2}} \approx 0.707V_m$
Rectification Efficiency (η)	40.6%	81.2%
Ripple Factor (γ)	1.21	0.482
Peak Inverse Voltage (PIV)	V_m	V_m (Bridge) vs $2V_m$ (Center-Tapped)
Output Ripple Frequency	f (50 Hz)	$2f$ (100 Hz)

2 Topic 1: P-N Junction Diode & Full-Wave Rectification

2.0.1 The P-N Junction & Shockley Diode Equation

Under thermal equilibrium, carrier diffusion creates a depletion region with a built-in potential barrier $V_0 \approx 0.7$ V (Silicon):

$$I = I_s\left(e^{\frac{V}{\eta V_T}} - 1\right)$$

Where thermal voltage $V_T = \frac{kT}{q} \approx 26$ mV at room temperature.

- **Forward Bias:** Applied voltage reduces the barrier, causing exponential diffusion current.
- **Reverse Bias:** Applied voltage widens the depletion layer; current is limited to reverse saturation current I_s ($\approx$ nA).

2.0.2 Full-Wave Bridge Rectifier Derivations

Using a bridge configuration of 4 diodes:

1. **Average DC Current:**

$$I_{\text{dc}} = \frac{1}{\pi}\int_0^{\pi} I_m \sin\theta d\theta = \frac{2I_m}{\pi} \approx 0.636 I_m$$

2. **RMS Current:**

$$I_{\text{rms}} = \sqrt{\frac{1}{\pi}\int_0^{\pi} (I_m \sin\theta)^2 d\theta} = \frac{I_m}{\sqrt{2}} \approx 0.707 I_m$$

3. **Rectification Efficiency (η):**

$$\eta = \frac{P_{\text{dc}}}{P_{\text{ac}}} = \frac{I_{\text{dc}}^2 R_L}{I_{\text{rms}}^2\left(R_L + 2R_f\right)} = \frac{\left(\left(2\frac{I_m}{\pi}\right)\right)^2 R_L}{\left(\left(\frac{I_m}{\sqrt{2}}\right)\right)^2 R_L} = \frac{8}{\pi^2} \approx 81.2\%$$

4. **Ripple Factor (γ):**

$$\gamma = \sqrt{\left(\frac{I_{\text{rms}}}{I_{\text{dc}}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} = \sqrt{1.2337 - 1} = 0.482$$

3 Topic 2: Zener Voltage Regulator & MOSFETs

3.0.1 Zener Diode Voltage Regulator Design

A Zener diode operated in its reverse breakdown region maintains a constant terminal voltage V_Z:

$$R_s = \frac{V_{\text{in,min}} - V_Z}{I_{\text{L,max}} + I_{\text{Z,min}}}$$

Power rating of Zener diode: $P_Z = V_Z I_{\text{Z,max}}$.

3.0.2 Operational Amplifiers (Op-Amps)

An ideal Op-Amp possesses infinite open-loop gain ($A_{\text{OL}} = \infty$), infinite input impedance ($R_{\text{in}} = \infty$), and zero output impedance ($R_{\text{out}} = 0$):

- **Inverting Amplifier:** $V_{\text{out}} = -\left(\frac{R_f}{R_1}\right)V_{\text{in}}$
- **Non-Inverting Amplifier:** $V_{\text{out}} = \left(1 + \frac{R_f}{R_1}\right)V_{\text{in}}$
- **Virtual Ground:** Due to infinite gain, $V_+ \approx V_-$.

4 Topic 3: Worked VIT FAT Question Bank Problems

Problem 6.1 (BEEE102L FAT April 2025 — 10 Marks): A Zener diode with breakdown voltage $V_Z = 10$ V and minimum Zener current $I_{\text{Z,min}} = 5$ mA regulates a supply voltage varying between 15 V and 25 V. If load resistance varies between 500Ω and 1000Ω, calculate: (i) Value of series resistor R_s, and (ii) Maximum power dissipated in the Zener diode.

Solution:

1. Load currents:

$$I_{\text{L,max}} = \frac{V_Z}{R_{\text{L,min}}} = \frac{10\text{ V}}{500\Omega} = 20\text{ mA}$$

$$I_{\text{L,min}} = \frac{V_Z}{R_{\text{L,max}}} = \frac{10\text{ V}}{1000\Omega} = 10\text{ mA}$$

2. Design series resistor R_s for worst-case minimum input voltage:

$$R_s = \frac{V_{\text{in,min}} - V_Z}{I_{\text{L,max}} + I_{\text{Z,min}}} = \frac{15 - 10}{20 \times 10^{-3} + 5 \times 10^{-3}} = \frac{5}{25 \times 10^{-3}} = 200\Omega$$

3. Maximum Zener current occurs at maximum input voltage and minimum load current:

$$I_{\text{total,max}} = \frac{V_{\text{in,max}} - V_Z}{R_s} = \frac{25 - 10}{200} = \frac{15}{200} = 75\text{ mA}$$

$$I_{\text{Z,max}} = I_{\text{total,max}} - I_{\text{L,min}} = 75\text{ mA} - 10\text{ mA} = 65\text{ mA}$$

4. Maximum power dissipated:

$$P_{\text{Z,max}} = V_Z I_{\text{Z,max}} = 10\text{ V} \times 65 \times 10^{-3}\text{ A} = 0.65\text{ W} = 650\text{ mW}$$

. **Answer:** $R_s = 200\Omega$, $P_{\text{Z,max}} = 0.65$ W.

5 Quick Self-Test Checklist Before Your Exam

1. Derive the ripple factor and efficiency of a full-wave bridge rectifier.
2. Why is PIV of a center-tapped full wave rectifier double that of a bridge rectifier?
3. Design a Zener diode voltage regulator given input voltage fluctuations and load range.
4. Derive the voltage gain of an inverting and non-inverting operational amplifier.